Class 6

Chemistry

Volume 2

Solutions

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3. CHANGES AROUND US

WORKSHEET – 01

Level – I

1) B 2) A 3) A 4) B 5) C 6) B

7) C 8) B 9) A 10) B 11) C

Level – II

1) B 2) A 3) D 4) B 5) C 6) C

7) B 8) D 9) D 10) B 11) D 12) C

13) B 14) C 15) B 16) D

Level – III

17) A, C 18) A,B,C 19) C 20) B 21) A 22) B

23) a-r,s;b-t;c-p,q,r,s;d-p 24) A,C,D 25) B,C,D 26) A 27) A

28) A 29) B 30) A,C,D 31) A,B,C,D

WORKSHEET - 02

Level – I

1) A 2) A 3) C 4) B 5) D 6) B

7) B 8) C

Level – II

1) D 2) B 3) A 4) C 5) A 6) D

7) B 8) B 9) B 10) C 11) B 12) A

13) C 14) A 15) A

Level – III

16) A,B 17) A,B,C,D 18) B 19) D 20) D 21) C

22) a-s;b-p;c-q;d-r 23) 6 24) A,B,D 25) A,B,C,D

26) A,B,C 27) A 28) A 29) C 30) C

31) a-t; b-p; c-q; d-r 32) 2 33) A,B,C,D 34) A,B,D

35) A,B,C,D 36) A 37) 26

4. ATOMIC STRUCTURE

WORKSHEET – 1

Level – I

1. 1	2. 2	3. 4	4. 2	5. 2	6. 3
7. 3	8. 4	9. 4	10. 2	11. 3	12. 3

Level – II

1. B	2. D	3. B	4. A	5. A	6. C
7. B	8. C	9. C	10. A	11. C	12.B

Level – III

13. A,B,C	14. A,C	15. A	16. C	17. a-s; b-p; c-r; d-q;e-t;
18. a-q; b-s; c-p; d-r		19. D	20. A	21. B	22. 1
23. C	24. B	25. B		

WORKSHEET – 2

Level – I

1. C	2. B	3. B	4. B	5. A	6. D
7. C	8. C				

Level – II

1. B	2. B	3. C	4. C	5. A	6. A
7. B	8. A	9. C	10. C	11. A	12. B

Level – III

13. A,B,C,D	14. C,D	15. a-t; b-r; c-s; d-q; e-p			
16. a-r; b-t; c-s; d-q;e-p;		17. A	18. C	19. A	20. B
21. C	22. 7	23. A	24. C	25. A	

WORKSHEET – 3

Level – I

1. B	2. A	3. D	4. A	5. C

Level – II

1. B	2. A	3. B	4. C	5. B	6. B
7. C	8. B	9. A	10. A	11. A	12. A
13. C					

Level – III

14. C	15. A,B,C	16. a-r; b-q; c-p; d-s;		17. A	18. A
19. A	20. C	21. C	22. C	23. B	24. D
25. C					

WORKSHEET – 4

Level – I

1. A 2. C 3. A 4. C 5. C

Level – II

1. A 2. B 3. C 4. A 5. B 6. B
7. D 8. A 9. C 10. C 11. C 12. A
13. C

Level – III

14. A,B,C 15. A,B,C,D 16. A 17. A
18. a-r,b-p,c-q, d-s 19. a-r, b-q, c-p, d-s 20. B 21. C
22. C 23. C 24. C 25. A

WORKSHEET – 5

Level – I

1. 3 2. 2 3. 2 4. 1 5. 2 6. 3
7. 4 8. 2 9.2 10.3

SOLUTIONS

2. Electrons enters into orbital which has lower $(n + \ell)$ value.
3. According to Moeller's chart.
4. 2 d^5 not possible, because n = 2 consists of only 2 s, 2p.
5. K = 2, L = 8, M = 6
   $1s^2\ 2s^2 2p^6\ 3s^2 3p^4$
7. $Z = 14 \Rightarrow 1s^2 2s^2 2p^6 3s^2 3p^2 3d^0$



8. $Z = 15 \Rightarrow 1s^2 2s^2 2p^6 3s^2 3p^3$

Level – II

1. 2 2. 1 3. 1 4. 3 5. 2 6. 4
7. 4 8. 2 9. 4 10. 3 11. 3 12. 3

1. According to Moeller's chart 4p < 5 s < 4 d < 5p ... .
3. $1\ s^2 2\ s^2 2p^1 3\ s^2 3p^3$ 
6. 6 s < 4f < 5 d < 6p <
7. $Z = 1\ s^2 2\ s^2 2p^6 \Rightarrow Ne, F^-, O^{-2} \rightarrow$ By gaining electron
   $\Rightarrow Na^+, Mg^{+2}, Al^{+3} \rightarrow$ By losing electron.
8. $Cu^{+2} \rightarrow 1\ s^2 2\ s^2 2p^6 3\ s^2 3p^6 4\ s^0 3\ d^9$
   $Fe^{+2} \rightarrow 1\ s^2 2\ s^2 2p^6 3\ s^2 3p^6 4\ s^0 3\ d^6$
   $Ni^{+2} \rightarrow 1\ s^2 2\ s^2 2p^6 3\ s^2 3p^6 4\ s^0 3\ d^8$

Level – III

13. A,B,C,D	14.A,B,C,D	15. A	16. D	17. C	
18. a-r; b-s; c-q; d-t; e-p		19. A	20. C	21. B	22. D
23. C	24. A	25. A	26. B		

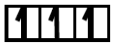
WORKSHEET – 6

Level – I

1. C	2. B	3. B	4. C	5. B	6. D
7. A	8. C	9. B	10. D		

Level – II

1. B	2. D	3. A	4. B	5. D	6. B
7. D	8. D	9. B	10. D	11. B	

6. $N \rightarrow 1\ s^2 2\ s^2 2p^3$ 

Level – III

12. A,B	13. A,C,D	14. a-r; b-p; c-q; d-t; e-s			15. A
16. C	17. A	18. D	19. B	20. C	21. B
22. D	23. B	24. B	25. A	26. A	

18. O shell $-2n^2 = 2(5)^2$
    $= 2 \times 25$
    $= 50$ electrons
20. $1s^2 2s^2 2p^6 3s^2 3p^5$
    $2 + 5 = 7$
22. $Z = 47 \Rightarrow [Kr]5\ s^1 4\ d^{10}$
    Number of 3 d electrons = 10
    Number of 4 d electrons = 10 = 20
25. $Z = 35$, Number of protons = 35
    $A = 80$, Number of neutrons $= 80 - 35 = 45$

5. PERIODIC CLASSIFICATION

WORISHEET – 1

Level – I

1) D 2) C 3) D 4) B 5) C 6) C
7) A

Level – II

1) A 2) C 3) C 4) C 5) A 6) D
7) C

Level – III

8) A,C 9) D 10) B 11) A 12) C
13) a-r;b-s;c-p;d-t 14) 23.0 15) A 16) A 17) B

WORKSHEET – 2

Level – I

1) 1 2) 1 3) 3 4) 2 5) 1 6) 4
7) 2

Level – II

1) D 2) A 3) A 4) C 5) B 6) D
7) B 8) A

Level – III

9) A,B,C,D 10) C 11) C 12) D 13) C
14) a-q; b-r; c-s; d-p 15) 35.5 16) A,C 17) A
18) a-q; b-s; c-p; d-r

WORKSHEET -3

Level – I

1) A 2) A 3) A 4) B 5) D 6) D
7) A 8) C

Level – II

1) A 2) A 3) B 4) D 5) A 6) C
7) A 8) A 9) B

Level – III

10) A,D 11) A 12) D 13) D 14) D
15) a-r; b-p; c-s; d-q 16) 69.9 17)A,B,C 18) C 19) A
20) A,D

WORKSHEET -4

Level – I

1) B 2) B 3) B 4) C 5) A 6) D
7) B

Level – II

1) C 2) B 3) C 4) B 5) A 6) B
7) B 8) A 9) C 10) C 11) A 12) D

Level – III

13) A,B,C,D 14) B 15) A 16) C
17) a - q, b -r, c -r, d -s 18) A,B,C,D 19) A 20) A,B,C

WORKSHEET – 5

Level – I

1) 2 2) 1 3) 2 4) 3 5) 3 6) 2
7) 2 8) 2 9) 4 10) 3

Level – II

1) B 2) B 3) D 4) A 5) B 6) A
7) C 8) D 9) B 10) B 11) D 12) B
13) C 14) D 15) A 16) C 17) A,B,C 18) A
19) A 20) C 21) a-s; b-r; c-q; d-p, t 22) 6 23) A,B,C
24) A,B,C

WORKSHEET – 6

Level – I

1) B 2) D 3) A 4) B 5) B

Level – II

1) B 2) A 3) A 4) B 5) D 6) D
7) B 8) D 9) D 10) B 11) C 12) A
13) A

Level – III

14) C 15) A, B 16) A,B,C 17) B 18) A 19) D
20) a - p, b -q, c -s, d -r 21) A,B,C,D 22) A,B
23) a -r; b -q; c -p; d -p

6. MOLE CONCEPT

WORKSHEET – 1

Level – I

1) A	2) B	3) B	4) A	5) C	6) B	7) A
8) A	9) C	10) B				

Level – II

1) C	2) D	3) D	4) A	5) D	6) B	7) B
8) A	9) B	10) C	11) B	12) B	13) B	14) A
15) C	16) B					

5. 1 gram molecule = 1 mole
2 gram molecule = 2 moles
1 mole of CH_4 = 16 g
2 moles of CH_4 = x
$x = 16 \times 2 = 32$

7. $CaCO_3$
$Ca - 40$
C -12
$O - 48$
So $CaCo_3$ molecular weight is 100 g

11. $O_3 = 48$ g
Atomic mass of oxygen = 16 g

12. N_2O
$N \rightarrow 2 \times 14 = 28$
$O \rightarrow 1 \times 16 = 16$
44

13. $Ca_3(PO_4)_2$
$Ca \rightarrow 3 \times 40 = 120$
$P \rightarrow 2 \times 31 = 62$
$O \rightarrow 8 \times 16 = 128$
310

14. Moleclelar weight of $C_{60}H_{122}$
$C \rightarrow 60 \times 12 = 720$
$H \rightarrow 122 \times 1 = 122$
842
6.023×10^{23} Molecuels $-842g$
1 Molecules $\rightarrow \times$
$x = \dfrac{842}{6.023 \times 10^{23}} = 1.397 \times 10^{-21}$

15. Atomic weight $'x' = \frac{(90\times200)+(8\times199)(2\times302)}{100}$
$= \dfrac{18000 + 1592 + 604}{100}$
$= \dfrac{20196}{100} = 201.96$
$= 202$

Level – II

17) A, B, C 18) A, B, C 19) a-p,q, b-q, c-r,t, d - s, t, e- s 20) A, B 21) B
22) C 23) B 24) a-q, b-s,q, c-p,d-r 25) A, B, C

21. $2H + 1\ S + 4O$
$= 2 + 32 + 4 \times 16$
$= 98$ g
22. $HCl = 36.5$ g
$CH_4 = 16$ g
$= 36.5 - 16 = 20.5$ g

WORKSHEET – 2

Level – I

1) A 2) B 3) B 4) D 5) A 6) B 7) B
8) B 9) D 10) B

9. 1 mole of electron $= 6.023 \times 10^{23}$
1 electron $= 9.1 \times 10^{-25}$ kg
6.023×10^{23} electrons = x
x $= 6.023 \times 10^{23} \times 9.1 \times 10^{-25} = 0.55$mg

Level – II

1) B 2) C 3) D 4) A 5) B 6) D 7) C
8) D 9) B 10) B 11) B 12) D 13) B 14) C
15) A

9. 36.5 g of HCl = 1 mole
73 g of HCl = 2 mole
10. 2 g of H_2 = 1 mole
1 mole contains 6.023×10^{23}
4 g of H_2 = 2 moles
$2 \times 6.023 \times 10^{23} = 12.046 \times 23$
11. 44 g of CO_2 = 22.4 lit
22 g of CO_2 = 11.2lit
14. $32g$ contains $\rightarrow N_A$ atomsof ' S '
$2.4g$ contains $\rightarrow x$
$x = \frac{2.4 \times N_A}{32}$
$= 0.075N_A$
15. Molar Mass $CO_2 = 44$
$CO_2 = 200\text{mg} = \frac{0.2}{44}$
$CO_2 4.55 \times 10^{-3}$ mol
no. of moleules of CO_2 removed $= 10^{21}$
1 mole $\rightarrow 6.023 \times 10^{23}$ molecules
$x \rightarrow 10^{21}$
$x = \frac{10^{21}}{6.023 \times 10^{23}}$

$x = 1.66 \times 10^{-3}$ mol
10_2 left $= (4.55 - 1.66)10^{-3}$
$= 2.89 \times 10^{-3}$ mol

16. no.of moles of 4 g of $H_2 = 2$
$\frac{4}{2} = 2$
no.of moles of 32 of 0_2
$\frac{32}{32} = 1$
$2:1$

Level – III

16) B	17) A,C,D	18) B	19) C	20) B	21) a-s,b-r, c-p,d-q
22) B, C	23) B, C	24) B,C,D	25) C	26) A	27) C
28) a-q,b-r,c-p,d-q		29) B,D	30) C	31) A	

22. 1 mole = 44 g of Carbondioxide
0.5 mole = 22 g of Carbondioxide
1 mole = 22.4 litres of CO_2
0.5 mole = 11.2 litres of CO_2

23. 28 g $N_2 \rightarrow 22.4l$
$56g \rightarrow x$
1 mole = 22.4 lit of N_2
2 mole = 44.8 lit of N_2
$x = \frac{22.4 \times 56}{28}$
$= 44.8l$

24. 1 mole = 22.4 litres of CH_4
0.5 mole = 11.2 litres of CH_4
$22.4l \rightarrow N_A$
$11.2l \rightarrow x$
$x = \frac{11.2 \times N_A}{22.4}$
$= \frac{N_A}{2}$
1 mole of methane = 16 g
0.5 mole of methane = 8 g

28. 22 g of H_2 = 1 mole
= 22.4lit
6 g of H_2 = 3 mole
= 67.2lit
8 g of 0_2 = 0.25 mole
= 5.6 lit
32 g of 0_2 = 1 mole
= 22.4 lit

29. 80 grams of Oxygen = 2.5 moles
1 mole = 32 g of Oxygen
2 g of Hydrogen = 1 mole

5 g of Hydrogen = 2.5 mole
1 mole = 44 g of CO_2
2.5 mole = 110 g of CO_2

30. $CaCo_3$
Total no. of Protons $= 20 + 6 + (3 \times 8) = 50$
100 g contains $\rightarrow 50 \times 6.02x \times 10^{23}$
log contains $\rightarrow x$

$$x = \frac{50 \times 6.023 \times 10^{23} \times 10}{100}$$

$$= 3.0115 \times 10^{24}$$

WORKSHEET – 3

Level – I

1) C 2) B 3) C 4) B 5) D 6) B 7) D
8) B

10. 1 moles of $CH_4 = 2 \times 12 + 4 \times 1 = 16$ g
2 moles of $CH_4 = 32$ g

Level – II

1) B 2) B 3) A 4) B 5) B 6) A 7) B
8) C 9) C 10) B 11) B

1. 1 mole $KMNO_4 = 1\ K + 1MN + 40 = 6\ N$
2. 28 g of N_2 molecule $\rightarrow$ 1 mole
56 g of N_2 Molecule $\rightarrow$ 2 mole
4. 1 Mole of $H_2SO_4 = 98$ g
x Mole of $H_2SO_4 = 294$ g

$$x = \frac{294}{98} = 3$$

5. 1 mole $= 6.023 \times 10^{23}$ molecule
$x = 6.023 \times 10^{21}$

$$\frac{6.023 \times 10^{21}}{6.023 \times 10^{23}} = 0.01 \text{ mole}$$

6. $O_2 : H_2 : CH_4$
$32 : 2 : 16$
$16 : 1 : 8$
9. 100 g of $CaCO_3 = 50$ protons
$= 50 \times 6.023 \times 10^{23}$
50 g of $CaCO_3 = x$

$$\frac{50 \times 50 \times 6.023 \times 10^{23}}{100}$$

$$= 1.50 \times 10^{25}$$

11. 2lit of water = 200gms
18 g $= 6.023 \times 10^{23}$
2000 g = x

$$x = \frac{6.023 \times 10^{23} \times 2000}{18}$$

$$x = \frac{6.023 \times 2 \times 10^{26}}{18} = 0.699 \times 10^{26}$$

Level – III

12) A, D 13) A 14) C 15) A 16) a - s, b - r, c -q, d -p

17) B, C, D 18) a - r, b - s, c - q, d – p 19) C, D 20) A, B, D

14. 16 g of CH_4 = 1 mole
8 g of CH_4 = 0.5 mole

16. $CO_2 = 12 + 32 = 44$ g
$H_2O = 2 + 16 = 18$ g
$Na_2CO_3 = 2 \times 23 + 12 + 48 = 106$ g
$C_2H_2 = 2 \times 12 + 2 = 26$ g

17. 16 gO_2 = 0.5 mole
22 gCO_2 = 0.5 mole
14 g N_2 = 0.5 mole
24 g of O_3 = 0.5 mole

WORKSHEET – 4

Level – I

1) B 2) B 3) B 4) B 5) B

4. Molecular mass = 2 × VD
= 2 × 14 = 28 g.

Level – II

1) B 2) A 3) B 4) D 5) A 6) C

7) C 8) D 9) B

1. VD = 16
Molecular mass = 2 × VD
= 2 × 16
= 32 g.

5. 22.4 lit of O_2 = 1 mole
56 lit of O_2 = x
$x = \frac{56}{22.4}$
= 2.5 moles

8. $3Bacl_2 + 2Na_3PO_4 \rightarrow Ba_3(PO_4)_2 + 6NaCl$
0.3
0.2
0.1
0.6

9. CH_4 = 16 g
VD = 16/2 = 8
N_2 = 28 g
VD = 28/2 = 14

$H_2O = 18$ g
$VD = 18/2 = 9.$
$CO_2 = 44$ g
$VD = 44/2 = 22.$

Level – III

10) A, B, D 11) A, D 12) D 13) C 14) C
15) a -q, b - r, c - s, d -p 16) C 17) A 18) B 19) A, B, C, D

16. Molecular mass = 2 × vapourdensity
$V.P = \frac{M \cdot M}{2} = — = 8.5$

18. 2 × Vapour Density
$= 2 \times 35.5$
$= 71$

WORKSHEET – 5

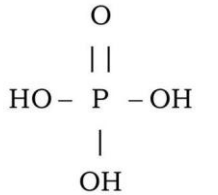
Level – I

1) C 2) A 3) C 4) C 5) B 6) A 7) C

1. Equivalent weight of $Na = \frac{\text{Atomicmass}}{\text{valency}} = \frac{23}{1} = 23$
2. Equivalent weight of $O = \frac{\text{Atomicmass}}{\text{valency}} = \frac{16}{2} = 8$

Level – II

1) 4 2) 1 3) 3 4) 3 5) 2 6) 2
7) 3 8) 2 9) 1 10) 2 11) 3 12) 3
13) 2 14) 2 15) 2

1. Equivalent weight of $He = \frac{\text{Atomicmass}}{\text{valency}} = \frac{4}{0}$ = Undefined
2. Equivalent weight of $HF = \frac{\text{Molecular Mass}}{\text{valency of anyion}}$
$= \frac{20}{1} = 20$
6. Equivalent weight of $HNO_3 = \frac{1+14+48}{1} = \frac{63}{1} = 63$
7. 
$H_3PO_4 in\ aqueous\ \ state\ \ releases\ 3H^+$
8. $Al(OH)_3$-it replaces 3 Hydroxyl ions
9. Na_2O
Atomic weight of sodium = 23 g
Equivalent weight = 23 g

12. Atomic weight of Oxygen is 16 g
Equivalent weight of Oxygen $= \frac{16}{2} = 8$

13. Molecular weight = 74 g
Equivalent weight $= \frac{74}{2} = 37.$

14. Equivalent weight of oxalic acid $= \frac{(2\times12)+(4\times16)+(2\times1)+(2\times18)}{2} = \frac{126}{2} = 63$

15. Equivalent weight = Atomic weight / Valency
Atomic weight $= 12 \times 2 = 24$ g

Level – III

16) A, B, C 17) C 18) B 19) C 20) a - r, b - q, c - p, d-s
21) A, B 22) a -s, b -r, c -s, d -p 23) A, C 24) a - q, b - p, c - r, d -s
25) C

19. Atomic weight of Aluminium = 27 g
Aluminium valency = 3
So it combines with 3 chlorine atoms its weight = 106.5 g
So equivalent weight of $x = 27/3 = 9$

20. Equivalent weight of Magnesium $= 24/2 = 12$
Equivalent weight of Iron in Ferrous compounds $= 56/2 = 28$
Equivalent weight of Iron in Ferric compounds $= 56/3 = 18.66$

WORKSHEET – 6

Level – I

1) A 2) B 3) B 4) D 5) D

4. Equivalent weight of $Na = \frac{\text{Atomicmass}}{\text{valency}} = \frac{23}{1} = 23$

5. $MgCl_2 = 24 + 71 = 95$
Equivalent weight $= \frac{\text{Molecular weight}}{\text{Charg e on cation (or)anion}}$
$= \frac{95}{2} = 47.5$

Level – II

1) D 2) A 3) A 4) B 5) C 6) D 7) A
8) B

2. Equivalent weight of $KMnO_4 = \frac{\text{molecular weight}}{\text{changein oxidation number}}$

$$2KMnO_4 + 3H_2SO_4 \rightarrow K_2SO_4 + 2MnSO_4 + 3H_2O + 5(O)$$
$$Mn^{+7}$$
$$= \frac{158}{5} = 31.6$$

3. $2KMnO_4 + H_2O \rightarrow 2KOH + 2MnO_2 + 3(O)$
Mn^{+7} Mn^{+4}

Equivalent weight $= \frac{\text{mol.weight}}{\text{Change in Oxidation number}} = \frac{158}{3} = 52.66$

4. $2KMnO_4 + 2KOH \rightarrow 2\ K_2MnO_4 + H_2O + (O)$

Mn^{+7}

Mn^{+6}

Equivalent weight $= \frac{\text{Mol.weight}}{\text{Change oxidation number}}$

$\frac{158}{1} = 158$

Level – III

9) A, B, C 10) B 11) B 12) B 13) a - q, b -r, c -s, d - p
14) A, B, C, D 15) A 16) C 17) B 18) B 19) B
20) A

11. Equivalent weight $= Mgcl_2 = \frac{24+71}{2} = \frac{95}{47.5} = 47.5$

Equivalent weight $= \frac{\text{atomic weight}}{\text{valency}}$

$12 = \frac{\text{atomic weight}}{2}$

atomic weight - 24

13. (a) $CaCl_2 = \frac{110}{2} = 55$

(b) $KOH = \frac{56}{1} = 56$

(c) Equivalent weight of $Al_2(SO_4)_3 = \frac{(2\times27)+(3\times32)+(12\times16)}{6} = \frac{342}{6} = 57$

(d) $CuSO_4 = \frac{160}{2} = 80$

WORKSHEET – 7

Level – I

1) C 2) D 3) D 4) D

Level – II

1) B 2) A 3) B 4) B 5) D 6) C 7) B

1. $\frac{(W_1)\text{ Weight of } Ag}{(W_2)\text{ Weight of } Zn} = \frac{\text{Equivalent weight of Ag}(E_1)}{\text{Equivalent weight of } Zn(E_2)}$

$\frac{6.9119}{2.08} = \frac{108}{E_2}$

$E_2 = \frac{108 \times 2.08}{6.9119} = 32.5$

2. $CuO + H_2 \rightarrow Cu + H_2O$

Wt of CuO − wt.of Cu = wt o.f O_2

$0.8567g - 0.6842g = 0.1725g$

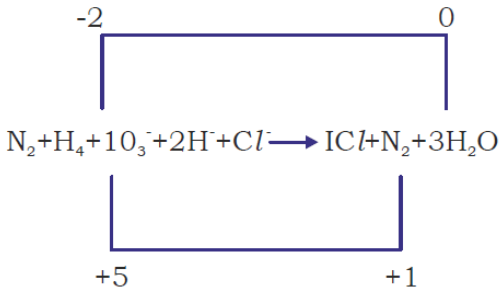
Oxygen Equivalent weight $(E_1) = 8$

$$\frac{\text{Weight of } O_2(w_1)}{\text{Weight of } Cu(w_2)} = \frac{\text{Equ.wt.of } O_2(E_1)}{\text{Equ.wt.of } Cu(E_2)}$$

$$\frac{0.1725}{0.6842} = \frac{8}{E_2}$$

$$E_2 = \frac{8 \times 0.6842}{0.1725} = 31.73$$

4.



Change in oxidation number of $N_2H_4(-2to0) = 2$

Equivalent weight $N_2H_4 = \frac{\text{Molecular weight}}{\text{Change in oxidation number}} = \frac{32}{2} = 16$ g

Change in oxidation number of $KIO_3 = (+2 + 0 + 1) = 4$

Equivalent weight $KIO_3 = \frac{\text{Molecular weight}}{\text{Changein oxidation number}} = \frac{214}{4} = 53.5$

6. Eqivalent weight of $H_3PO_2 = \frac{\text{Molecular weight of } H_3PO_2}{\text{Basicity}} = \frac{M}{1} = M$

7. Weight of Iron $w_1 = 27.93g$

Weight of Oxygen $w_2 = ?x$

Equivalent weight of Iron $E_1 = 18.62$

Equivalent weight of Oxygen $E_2 = 8$

As per law of gram equivalents

$$\frac{w_1}{E_1} = \frac{w_2}{E_2}$$

$$\frac{7.93}{18.62} = \frac{x}{8}$$

$$x = \frac{27.93 \times 8}{18.62}$$

$= 12$ g